The zero-phase dictionary and the constant cross-validation

Claude, with Daniel Murfet

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Abstract

Astra #18's "dictionary": the two-dimensional quadratic chart of the random-phase Taylor tree, at zero phase $\xi \equiv 0$ and unit amplitude $\eta \equiv 1$ (Taylor data $x = 0$, $y = \delta_{00}$), is *exactly* the unit-box monomial integral at parameter N^2 (`twoDAmp_zeroPhase_eq_monomial_sq`). A square-parameter transport lemma turns a power–log limit in T into one in $N = \sqrt{T}$ with the logarithmic coefficient multiplied by 2^r (`tendsto_powLog_comp_sq`). By uniqueness of limits the Taylor-tree leading log coefficient at zero phase equals *twice* the monomial mixed constant at $\lambda = p/2$ (`canonA_zeroPhase_eq_two_mul_monomialMixedConst`):

$$\frac{y_{00}}{k_1 k_2} \int_0^\infty s^{p-1} e^{-\beta s^2}\, ds = \frac{y_{00}\, \Gamma(p/2)\beta^{-p/2}}{2 k_1 k_2} = 2 \cdot \frac{\Gamma(p/2)\beta^{-p/2}}{1! \cdot 2k_1 \cdot 2k_2},$$

the factor 2 being $\log N^2 = 2 \log N$. This cross-validates the constants of the two programmes at the level of theorems. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem ampCoeff_zero_delta ( ) (k) (s) : ampCoeff  (fun _ => 0) delta k s = delta k
theorem anaAmp_ampCoeff_zero_delta (  b u v s) :
    anaAmp (ampCoeff  (fun _ => 0) delta) b u v s = 1
theorem twoDAmp_const_eq_monomialCutoff (  b N) (h  h  k  k ) :
    twoDAmp  b N h  h  k  k  (fun _ _ _ => 1)
      = monomialBoxRealCutoff 2 ![h , h ] ![k , k ] b   (N ^ 2)
theorem twoDAmp_zeroPhase_eq_monomial_sq (  N) (h  h  k  k ) :
    twoDAmp  1 N h  h  k  k  (anaAmp (ampCoeff  (fun _ => 0) delta) 1)
      = monomialBoxReal 2 ![h , h ] ![k , k ]   (N ^ 2)
theorem tendsto_powLog_comp_sq (F) (l C) (r)
    (hF : Tendsto (fun T => F T / (T ^ (-l) * log T ^ r)) atTop (  C)) :
    Tendsto (fun N => F (N ^ 2) / (N ^ (-(2 * l)) * log N ^ r)) atTop (  (2 ^ r * C))
theorem canonA_zeroPhase_eq_two_mul_monomialMixedConst ( ) (h  h  k  k ) (h ) (hk ) (hk ) (p)
    (hp  : ((h  :  ) + 1) / k  = p) (hp  : ((h  :  ) + 1) / k  = p) :
    canonA   h  h  k  k  (fun i j s => ampCoeff  (fun _ => 0) delta (i, j) s) p
      = 2 * monomialMixedConst ![h , h ] ![k , k ] (p / 2)
```

2 Mathematics

The exponential coefficient array of the zero phase is δ_{00} (only the $n = 0$ term of the exponential series survives), so the amplitude array is δ_{00} and its analytic amplitude is the constant 1; the quadratic kernel $e^{-\beta(N u^{k_1} v^{k_2})^2}$ is $e^{-\beta N^2 u^{2k_1} v^{2k_2}}$, and two Bochner peels identify the iterated integral with the $\mathrm{Fin}\,2$ box integral. The Taylor tree gives twoDAmp $\sim A\, N^{-p} \log N$ with

$A > 0$ (units 154, 167); the monomial theorem at $T = N^2$ gives the same ratio with limit $2 \cdot \Gamma(p/2)\beta^{-p/2}/(2k_1 \cdot 2k_2)$. Consistency with the fluctuation-function normalisation: $J_p(0) = \frac{1}{2}\beta^{-p/2}\Gamma(p/2)$ (`gaussMomentJ_zero`), $S_{p/2}(a) = 2J_p(a)$, $A_p = y_{00}J_p(x_{00})/(k_1k_2)$.

3 Paper-facing meaning

The two independent developments — the random-phase Taylor tree in the variable $N = \sqrt{n}$ and the monomial normal-moment programme in the variable $N = n$ — agree on the leading constant where they overlap. In the paper's variable n: $Q(\sqrt{n})/(n^{-p/2}\log n) \to y_{00}\Gamma(p/2)\beta^{-p/2}/(4k_1k_2)$, i.e. the coefficient of $N^{-p}\log N$ is twice the coefficient of $n^{-p/2}\log n$.

4 Lean notes

- `isEquivalent_iff_tendsto_one` now takes `  x, v x   0` (not the older implication form).
- Indices of `Fin 2`: `Fin.cons u (Fin.cons v _) 1` needs `← Fin.succ_zero_eq_one, Fin.cons_succ, Fin.cons_zero`; `Matrix.cons_val_one` handles `![h , h ] 1`.
- Continuity of `fun w => Fin.cons u w` by `continuous_pi` and `Fin.cases`.
- A `simp` on a sum of indicator-like `if`s over `Fin 2` may rewrite it into a `Finset.card` of a set-builder; use `Fin.sum_univ_two` and `if_pos` instead.